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Electronic Circuit Design

Lecture notes

Chapter 9: Analog-to-Digital Conversion

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9 Analog-to-Digital Conversion

9.1 Sampling and Aliasing

An analog-to-digital converter (ADC) does two things: it samples the signal in time, and it quantizes each sample in amplitude. Both steps lose information, and both losses can be controlled by design. This section deals with sampling: how fast must we sample, and what happens if the signal contains frequencies that are too high for the sampling rate? The answer is the sampling theorem of Nyquist and Shannon [10]. Its practical consequence is a filter that every sampled system needs in front of the S&H stage: the anti-aliasing low-pass.

The sampling theorem

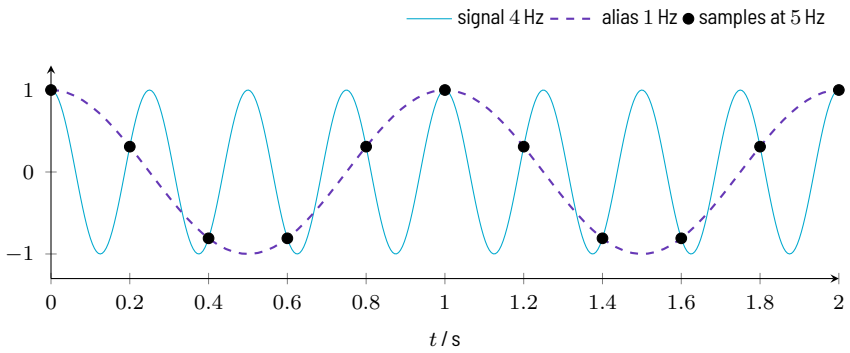
- a signal is sampled at the instants $t_k = k\,T_s$ with the sampling frequency $f_s = 1/T_s$

> **Sampling theorem (Nyquist–Shannon)**
> A signal that contains no frequencies at or above $f_s/2$ is completely determined by its samples taken at the rate f_s. $f_s/2$ is called the **Nyquist frequency**.

- the condition is $f_{\text{max}} < f_s/2$, i.e. more than two samples per period of the fastest component
- it applies to **every** component of the input, also to noise and interference
- capstone: $f_s = 5$ Hz $\Rightarrow$ only components below 2.5 Hz are allowed

The theorem says that the original signal can be reconstructed from its samples, provided that the band limit is respected. If the condition is violated, the samples are still taken, but they are the samples of a different, lower frequency. No processing after the sampler can tell the two cases apart.

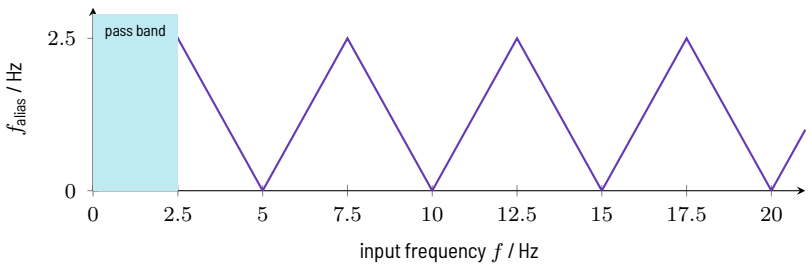
Aliasing



- a 4 Hz cosine sampled at 5 Hz gives the same samples as a 1 Hz cosine

> **Alias frequency**
> $f_{\text{alias}} = |f - k\,f_s|$, with the integer k that makes $f_{\text{alias}} \le f_s/2$

Folding



- every frequency is folded into the range $0 \ldots f_s/2$
- $f_s = 5$ Hz: 4 Hz $\to 1$ Hz; 12 Hz $\to 2$ Hz; 49 Hz $\to 1$ Hz; 50 Hz $\to 0$ Hz
- after sampling an alias cannot be distinguished from a real signal component

The 50 Hz problem

- mains interference (50 Hz) is coupled into every sensor cable
- capstone: $f_s = 5$ Hz $\Rightarrow f_{\text{alias}} = |50\text{ Hz} - 10 \cdot 5\text{ Hz}| = 0$ Hz
  - every sample sees the 50 Hz sine at the same phase
  - the result is a **constant DC offset** that biases every reading
  - its size depends on the (random) phase: anywhere between $-\hat{V}$ and $+\hat{V}$
  - nothing on the display looks wrong: the error is invisible
- with $f_s = 4.9$ Hz the alias is $|50\text{ Hz} - 10 \cdot 4.9\text{ Hz}| = 1$ Hz: a slow beat
- **the anti-aliasing low-pass in front of the S&H stage is mandatory**

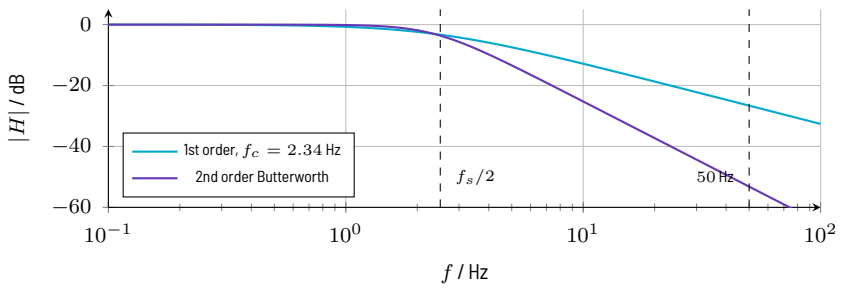
Because f_s is an integer fraction of the mains frequency, the interference is sampled at the same point of its period every time. The digital system sees a perfectly stable value, shifted by an unknown amount. With a slightly different sampling rate the same interference would show up as a slow oscillation of the readings, which is at least visible. The only remedy is to remove the interference before sampling.

Falstad: sampling 50 Hz hum

- input: 2 V DC (the wanted signal) plus 0.5 V at 50 Hz
- two S&H stages ($R = 1\,\text{k}\Omega$, $C_H = 100$ nF, track window 1 ms)
  - $f_s = 5$ Hz: every sample is 2.14 V: an offset of 0.14 V
  - $f_s = 4.9$ Hz: 2.14 V, 2.50 V, 2.14 V, 1.58 V, ...: a 1 Hz sine
- change the phase of the 50 Hz source: the offset at 5 Hz changes between 1.5 V and 2.5 V

▶ Falstad: `examples/adc/alias50.txt` The track window is only 1 ms here, so that the hold capacitor can follow the 50 Hz signal ($\tau = 0.1$ ms). The held value is the input at the end of the window, $t = 1$ ms: $2\text{ V} + 0.5\text{ V} \cdot \sin(2\pi \cdot 50\text{ Hz} \cdot 1\text{ ms} - 1.8°) = 2.14\text{ V}$, where $1.8°$ is the phase lag of the RC charging path.

Anti-aliasing filter



- pass the wanted signal, attenuate everything above $f_s/2$ below the resolution
- at 50 Hz: 1st order -26.6 dB (factor 21), 2nd order -53 dB (factor 460)
- the filter must be **before** the S&H: after sampling the damage is done

Design card: Nyquist & Anti-aliasing

- sampling theorem: all input components must be below $f_s/2$ (Nyquist frequency)
- alias: $f_{\text{alias}} = |f - kf_s|$, nearest multiple kf_s; result lies in $0 \ldots f_s/2$
- capstone: $f_s = 5$ Hz, 50 Hz hum $\rightarrow |50 - 10 \cdot 5|$ Hz $= 0$ Hz: constant offset on every reading
- low-pass before the S&H: $f_c \approx f_s/2$ (capstone 2.5 Hz), e.g. RC: $f_c = 1/(2\pi RC)$
- required attenuation: interference amplitude $\times |H(f)| < \frac{1}{2}$ LSB (better $\frac{1}{4}$ LSB)
- 1st order: $|H| \approx f_c/f$ (-20 dB/decade), 2nd order: $(f_c/f)^2$ (-40 dB/decade)
- higher f_s (oversampling) relaxes the filter: more distance between signal band and $f_s/2$

9.2 Quantization

The second step is quantization: the continuous voltage range is divided into 2^N intervals of equal width, and each sample is replaced by the number of its interval. The width of the interval is the smallest voltage step the converter can resolve; it is called LSB (least significant bit), because a change of the input by one LSB changes the least significant bit of the output code [6, 2].

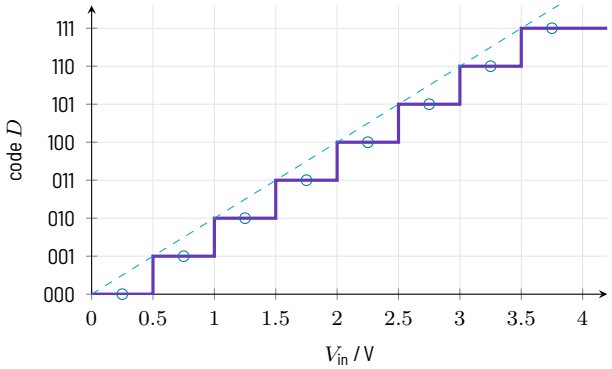
Resolution and LSB

LSB

$\text{LSB} = \dfrac{V_{\text{FS}}}{2^N}$ (V_{FS}: full-scale range, N: number of bits)

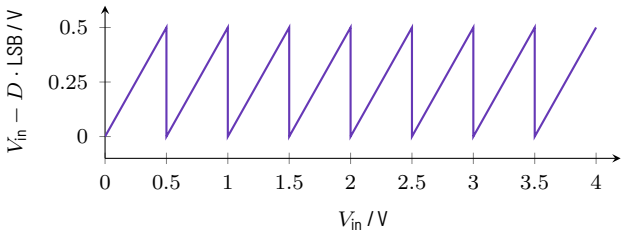
- capstone: $N = 3$, $V_{\text{FS}} = 4$ V: LSB $= 0.5$ V, 8 codes 000 …111
- 10 bit, 3.3 V (microcontroller): 3.2 mV; 16 bit, 5 V: 76 µV
- code for an input voltage (our flash converter):
  $D = \left\lfloor \dfrac{V_{\text{in}}}{\text{LSB}} \right\rfloor$, limited to $0 \ldots 2^N - 1$
- back to a voltage: $V_D = D \cdot$ LSB (lower edge) or $(D + \frac{1}{2})$ LSB (bin centre)

Transfer characteristic



- staircase: thresholds at $k \cdot$ LSB ($k = 1 \ldots 7$), the code changes at 0.5 V, 1.0 V, …, 3.5 V
- circles: bin centres 0.25 V, 0.75 V, …, 3.75 V: safe test points

Quantization error



- with thresholds at $k \cdot$ LSB: error $0 \ldots 1$ LSB (truncation), mean $\frac{1}{2}$ LSB
- shift all thresholds by $\frac{1}{2}$ LSB (ladder: $R/2$ at the bottom, $3R/2$ at the top): error $\pm\frac{1}{2}$ LSB
- or read the code as the bin centre $(D + \frac{1}{2})$ LSB: also $\pm\frac{1}{2}$ LSB

The two variants differ only by a constant offset of half an LSB. The capstone uses the simple ladder of eight equal resistors; its thresholds are at integer multiples of 0.5 V. A digital value D then means "the input is between $D \cdot 0.5$ V and $(D + 1) \cdot 0.5$ V".

Quantization noise and SNR

- for a busy signal the error behaves like noise, uniformly distributed over one LSB

  - rms value $V_{q,\text{rms}} = \dfrac{\text{LSB}}{\sqrt{12}}$
- full-scale sine: $V_{\text{rms}} = \frac{V_{\text{FS}}}{2\sqrt{2}} = \frac{2^N\,\text{LSB}}{2\sqrt{2}}$

Signal-to-noise ratio of an ideal N-bit converter

$$\text{SNR} = 20\log_{10}\frac{2^N\sqrt{12}}{2\sqrt{2}}\,\text{dB} = 6.02\,N + 1.76\,\text{dB}$$

- 3 bit: 19.8 dB; 8 bit: 49.9 dB; 12 bit: 74.0 dB; 16 bit: 98.1 dB
- every additional bit: 6 dB more (factor 2 in amplitude)

Falstad: quantization made visible

- 3-bit flash converter (next sections) followed by a 3-bit R-2R DAC
  - the DAC turns the code back into $V_q = D \cdot 0.5$ V
- input: triangle 0 V … 4 V, 0.25 Hz
- the scope shows V_{in} and the staircase V_q; the difference is always between 0 V and 0.5 V
- example readings: $V_{\text{in}} = 1.8$ V $\rightarrow V_q = 1.5$ V; $V_{\text{in}} = 3.98$ V $\rightarrow V_q = 3.5$ V

▶ Falstad: `examples/adc/quant_error.txt`

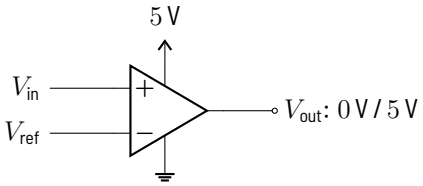
Design card: Reference Ladder & Quantization

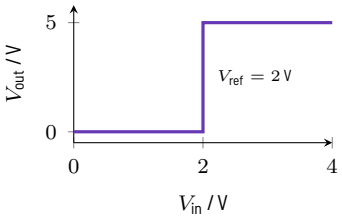
- LSB $= V_{\text{FS}}/2^N$; capstone: $4\,\text{V}/8 = 0.5$ V
- ladder of 2^N equal resistors from $V_{\text{ref}} = V_{\text{FS}}$ to ground: taps $V_k = k \cdot V_{\text{ref}}/2^N$, $k = 1 \ldots 2^N - 1$
- capstone: 8×1 kΩ from 4 V: taps 0.5 V, 1.0 V, …, 3.5 V; $I = 0.5$ mA
- code $D = \lfloor V_{\text{in}}/\text{LSB} \rfloor$; quantization error $0 \ldots 1$ LSB; with $R/2$ at the bottom and $3R/2$ at the top: $\pm\frac{1}{2}$ LSB
- test points: bin centres $(D + \frac{1}{2})$ LSB = 0.25 V, 0.75 V, …, 3.75 V
- ladder current $\gg$ comparator input currents; tap resistance $\leq R_{\text{total}}/4 = 2$ kΩ
- reference: stiff voltage source (reference IC + buffer); its error scales all thresholds
- SNR of an ideal converter: $6.02\,N + 1.76$ dB

9.3 The Comparator

The comparator is the 1-bit converter: it decides whether its input is above or below a threshold. Every converter contains at least one. An op-amp without feedback is a comparator, because its huge open-loop gain drives the output into one of the rails as soon as the input difference is more than a few microvolts. Dedicated comparator ICs are faster and have logic-compatible outputs [1, 2].

Open-loop op-amp as comparator





- $V_{out} = V_{OH}$ if $V_{in} > V_{ref}$, otherwise V_{OL}
- single 5 V supply: output levels fit 5 V logic directly (rail-to-rail output)
- Falstad: op-amp with *max output* 5 V, *min output* 0 V

Real comparators

- op-amps are slow when used open loop: recovery from saturation, slew rate (µs)
- comparator ICs (LM393, LM339, TLV3501): fast, no frequency compensation, logic outputs
  - LM393/LM339: **open-collector** output, needs a pull-up resistor (e.g. 4.7 kΩ to 5 V)
- input offset voltage (mV): shifts the threshold; must be $\ll$ LSB
- input bias current: flows through the ladder resistors, small threshold shift
- never use a comparator IC as an op-amp: it has no compensation and oscillates with feedback

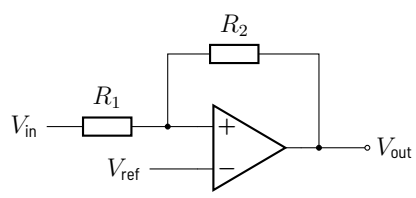
For the 3-bit capstone with an LSB of 0.5 V, any op-amp or comparator is good enough: offsets of a few millivolts are three orders of magnitude below the LSB. For an 8-bit flash converter with 4 V full scale (LSB $= 15.6$ mV) the offset of each comparator already matters.

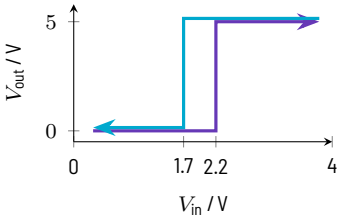
Chatter at slow, noisy inputs

- a slowly changing input with noise crosses the threshold several times
- the output switches back and forth (**chatter**): a counter counts wrong, an LED flickers
- remedy: **hysteresis**: two thresholds, V_{T+} for rising and V_{T-} for falling inputs
  - positive feedback (Schmitt trigger)
  - the hysteresis $V_{T+} - V_{T-}$ must be larger than the peak-to-peak noise
- Falstad: triangle 0 V … 4 V at 1 Hz plus 0.2 V at 50 Hz
  - plain comparator: 10 edges in one period; Schmitt trigger: exactly 2

▶ Falstad: `examples/adc/comparator_hyst.txt`

Schmitt trigger (non-inverting)





- switching when $V_+ = V_{ref}$, with $V_+ = V_{in}\frac{R_2}{R_1+R_2} + V_{out}\frac{R_1}{R_1+R_2}$:
  $V_T = V_{ref}\left(1 + \frac{R_1}{R_2}\right) - V_{out}\frac{R_1}{R_2}$

- V_{T+} (output low): $V_{\text{ref}}(1 + R_1/R_2)$; V_{T-} (output high): $V_{T+} - V_{\text{OH}} R_1/R_2$
- $R_1 = 10\,\text{k}\Omega$, $R_2 = 100\,\text{k}\Omega$, $V_{\text{ref}} = 2\,\text{V}$: $V_{T+} = 2.2\,\text{V}$, $V_{T-} = 1.7\,\text{V}$, hysteresis $0.5\,\text{V}$

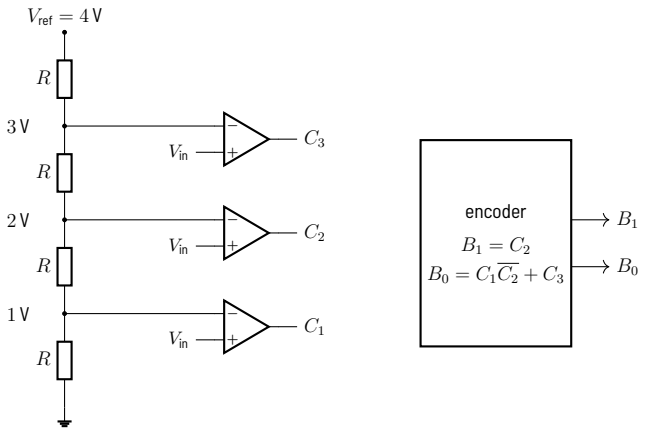
Design card: Comparator

- $V_{\text{out}} =$ high if $V_+ > V_-$, else low; op-amp open loop or comparator IC
- logic compatible: supply $0/5\,\text{V}$, rail-to-rail output (or open collector + pull-up $4.7\,\text{k}\Omega$)
- Falstad: op-amp, max output $5\,\text{V}$, min output $0\,\text{V}$; logic inputs switch at $2.5\,\text{V}$
- offset voltage $\ll$ LSB; input current $\times$ source resistance $\ll$ LSB
- hysteresis against chatter: R_1 from V_{in} to V_+, R_2 from V_{out} to V_+, V_{ref} at V_-
- $V_{T+} = V_{\text{ref}}(1+R_1/R_2)$, $V_{T-} = V_{T+} - V_{\text{OH}} R_1/R_2$, hysteresis $V_{\text{OH}} R_1/R_2 >$ noise (peak-to-peak)
- in a flash ADC behind an S&H: no hysteresis needed (the held input is constant), hysteresis would shift the thresholds

9.4 The Flash Converter

The flash (or parallel) converter is the most direct way to convert a voltage: compare it with all thresholds at the same time. An N-bit flash converter needs $2^N - 1$ comparators and a reference ladder that provides the thresholds. The comparator outputs form a thermometer code, which a small logic circuit turns into a binary number. The conversion takes only the delay of one comparator and a few gates, which makes flash converters the fastest of all architectures. The price is the exponential growth of the number of comparators: 7 for 3 bit, 255 for 8 bit.

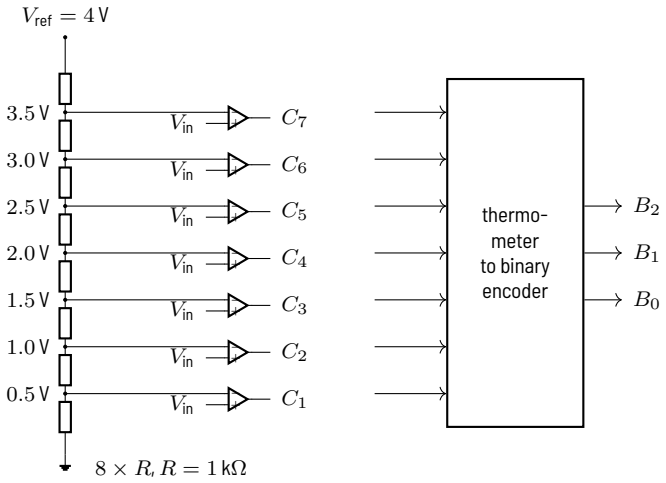
2-bit flash converter



- LSB $= 1$ V; thresholds 1 V, 2 V, 3 V; 3 comparators
- $C_k = 1$ if $V_{in} > k \cdot$ LSB

▶ Falstad: `examples/adc/flash2.txt`

3-bit flash converter (capstone)



- 7 comparators, $C_k = (V_{in} > k \cdot 0.5\,V)$; ladder current $4\,V / 8\,k\Omega = 0.5$ mA

Thermometer code

V_{in} / V	D	C_7	C_6	C_5	C_4	C_3	C_2	C_1	B_2	B_1	B_0
0.00 …0.50	0	0	0	0	0	0	0	0	0	0	0
0.50 …1.00	1	0	0	0	0	0	0	1	0	0	1
1.00 …1.50	2	0	0	0	0	0	1	1	0	1	0
1.50 …2.00	3	0	0	0	0	1	1	1	0	1	1
2.00 …2.50	4	0	0	0	1	1	1	1	1	0	0
2.50 …3.00	5	0	0	1	1	1	1	1	1	0	1
3.00 …3.50	6	0	1	1	1	1	1	1	1	1	0
3.50 …4.00	7	1	1	1	1	1	1	1	1	1	1

- like the column of a thermometer: all comparators below the input level are 1

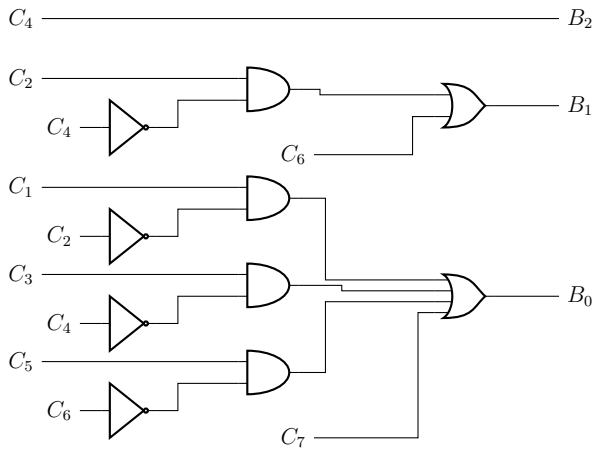
- the number of ones is the code D; only 2^N of the 2^{2^N-1} patterns are valid

Deriving the encoder

- the top of the column: $H_k = C_k\,\overline{C_{k+1}}$ is 1 only for $D = k$ (one-hot code, $C_8 := 0$)
- each output bit is the OR of the H_k whose code has this bit set:
  - $B_2 = H_4 + H_5 + H_6 + H_7 = C_4$
  - $B_1 = H_2 + H_3 + H_6 + H_7 = C_2\overline{C_3} + C_3\overline{C_4} + C_6\overline{C_7} + C_7 = C_2\overline{C_4} + C_6$
  - $B_0 = H_1 + H_3 + H_5 + H_7 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7$
- simplification uses the thermometer property: $C_{k+1} = 1 \Rightarrow C_k = 1$
  - $C_2\overline{C_3} + C_3\overline{C_4} = C_2\overline{C_4}$: "between level 2 and level 4"
  - B_2: four ones in a row means $D \geq 4$

The derivation can also be read directly from the table: B_2 is 1 for the lower half of the table, exactly where C_4 is 1. B_1 is 1 for the codes 2, 3 and 6, 7: the input is above level 2 but not above level 4, or it is above level 6. B_0 alternates with every code; it is 1 if the top of the thermometer column is at an odd position. The equations need three inverters, four 2-input AND gates, one 2-input OR gate and one 4-input OR gate.

Encoder logic



- 3 NOT, 4 AND, 2 OR gates: the same gates as in the Falstad model

Falstad: 3-bit flash converter with display

- input: slider 0 V … 4 V (label `Vin`); ladder 8×1 kΩ, 4 V
- comparator outputs and encoder signals are labeled nodes `C1`…`C7`, `nC2`, `a24`, …, `B0`…`B2`
- 7 LEDs as bar graph, 3 LEDs for $B_2B_1B_0$, each with $330\,\Omega$
- checked at the bin centres 0.25 V, 0.75 V, …, 3.75 V: codes 000, 001, …, 111
- default 2.6 V: five bar LEDs on, binary 101

▶ Falstad: `examples/adc/flash3.txt` Labeled nodes keep the drawing readable: every comparator output gets a label, and the gate inputs carry the same label names instead of long wires. Hover over a label to see its voltage. The tap voltages of the ladder are 0.5 V …3.5 V, the ladder current is 0.50 mA.

Priority encoder and bubble errors

- **bubble**: a wrong comparator decision inside the column, e.g. by offset or noise near a threshold

    - true code 3: 0000111; with bubble at C_2: 0000101
    - our encoder: $B_0 = C_1\overline{C_2} = 1$, $B_1 = 0$, $B_2 = 0 \Rightarrow$ 001: error of 2 LSB
- **priority encoder** (74HC148, 8 inputs $\rightarrow$ 3 bit): outputs the number of the **highest** active input
    - ignores everything below the top: a bubble does no harm
- bubble correction with 3-input gates: $H_k = C_{k-1}\, C_k\, \overline{C_{k+1}}$
- large flash converters: ones counter (Wallace tree) or Gray-code encoding

In the capstone the comparators decide on a held, constant voltage with an LSB of 0.5 V, so bubbles do not occur. In fast converters with hundreds of comparators they are a real problem, because the comparators near the input level see differences of only microvolts and decide at slightly different times.

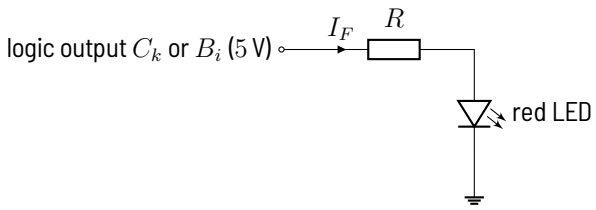
Design card: Thermometer-to-Binary Encoder

- thermometer code: $C_k = 1$ for all $k \le D$; the number of ones is the code D
- one-hot: $H_k = C_k\overline{C_{k+1}}$ (top of the column); $B_i =$ OR of all H_k with bit i of k set
- 3 bit (capstone): $B_2 = C_4 \quad B_1 = C_2\overline{C_4} + C_6 \quad B_0 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7$
- gates: 3 NOT ($\overline{C_2}, \overline{C_4}, \overline{C_6}$), 4 AND2, 1 OR2, 1 OR4
- check: sweep V_{in} over the bin centres 0.25 V … 3.75 V in steps of 0.5 V: 000 …111
- alternative: priority encoder (74HC148, active-low inputs and outputs), robust against bubbles

9.5 Displaying the Result

In the lab the digital value is shown on LEDs: a bar graph driven directly by the thermometer code, and three LEDs for the binary code. The bar graph is easy to read without any decoding; the binary LEDs show that the encoder works. Every LED is driven by a logic output (or comparator output) through its own current-limiting resistor.

LED with series resistor



Series resistor

$$R = \frac{V_{OH} - V_F}{I_F} = \frac{5\,\text{V} - 2\,\text{V}}{9\,\text{mA}} = 333\,\Omega \quad \rightarrow \quad \text{E12: } 330\,\Omega,\ I_F = 9.1\,\text{mA}$$

- red LED: $V_F \approx 1.8\,\text{V} \ldots 2.0\,\text{V}$; bright enough at $5\,\text{mA} \ldots 10\,\text{mA}$
- **one resistor per LED**: with a common resistor the current would depend on the number of LEDs that are on
- Falstad LED model: $V_F = 1.77\,\text{V}$, $I_F = 9.8\,\text{mA}$ with $330\,\Omega$

Current budget

- code 7: 7 bar LEDs + 3 binary LEDs on: $10 \times 9.1\,\text{mA} \approx 91\,\text{mA}$ from the $5\,\text{V}$ supply
- each output must deliver its LED current at a high level close to $5\,\text{V}$
  - in our circuit C_4 drives its bar LED, the B_2 LED and the gate inputs: $18\,\text{mA}$
  - 74HC gates: $\pm 25\,\text{mA}$ absolute maximum per output, V_{OH} drops under load
- better: buffer gates, low-current LEDs ($2\,\text{mA}$), or a driver IC (74HC245; ULN2003 sinks)
- LED to ground (output sources current) or LED to $5\,\text{V}$ (output sinks, active low): both are fine

The Falstad model has ideal outputs, so it does not show any of these loading effects. In hardware, look up the guaranteed output voltage at the required current in the data sheet. For a CMOS gate supplied with $5\,\text{V}$, V_{OH} is guaranteed above about $3.8\,\text{V}$ only up to a few milliamperes; at higher currents the LED current decreases accordingly.

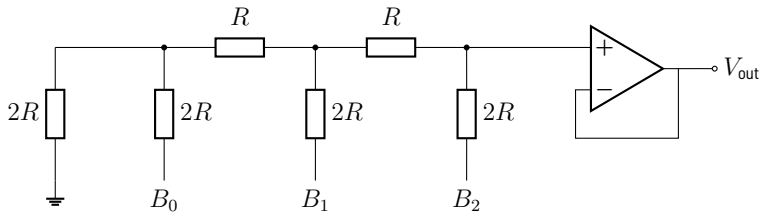
Design card: Falstad digital cheat-sheet

- gates: *Draw* → *Logic Gates, Input and Output*: AND, OR, NAND, NOR, XOR, Inverter; number of inputs and high level ($5\,\text{V}$) in the edit dialog (double click)
- logic levels: input high above $V_{high}/2 = 2.5\,\text{V}$; outputs are ideal $0\,\text{V}/5\,\text{V}$ sources
- comparator: op-amp with max output $5\,\text{V}$ and min output $0\,\text{V}$; inputs $-$ and $+$ as drawn
- *Logic Input* (click to toggle) and *Logic Output* (shows H/L) for quick tests
- *Labeled Node*: nodes with the same name are connected; use them for $C_1 \ldots C_7$, $B_0 \ldots B_2$; hover to read the voltage
- input voltage: DC voltage source or rail with a slider (edit dialog); clock: rail with square wave, frequency, duty cycle
- analog switch: closed if control $> 2.5\,\text{V}$, $R_{on} = 20\,\Omega$
- LED: always with a series resistor ($330\,\Omega$ at $5\,\text{V}$); lights above a few mA
- scope: right click on a node or element → *View in Scope*; set the time step to $0.1\,\text{ms}$ for $5\,\text{Hz}$ clocks

9.6 Other Converter Architectures

The flash converter is fast but expensive in comparators. Other architectures trade speed for resolution. Most of them contain a digital-to-analog converter (DAC) in a feedback loop, so we start with the simplest DAC, the R-2R ladder. Then we look at the successive-approximation converter found in every microcontroller, the dual-slope converter of digital multimeters and the sigma-delta converter used for audio and precision measurement [6, 7].

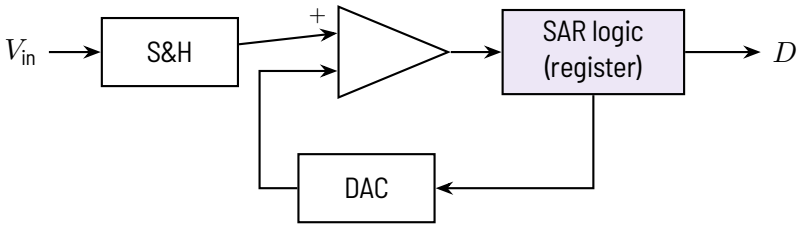
R-2R ladder DAC



- bit inputs: logic levels 0 V or V_{ref}; output resistance of the ladder: always R
- $V_{out} = V_{ref}\left(\frac{B_2}{2} + \frac{B_1}{4} + \frac{B_0}{8}\right) = V_{ref}\frac{D}{2^N}$
- only two resistor values, any number of bits: easy to integrate
- Falstad: $R = 10\,\text{k}\Omega$, 5 V logic, counter at 1 Hz/0.5 Hz/0.25 Hz: staircase in steps of 0.625 V, $D = 5$: 3.125 V

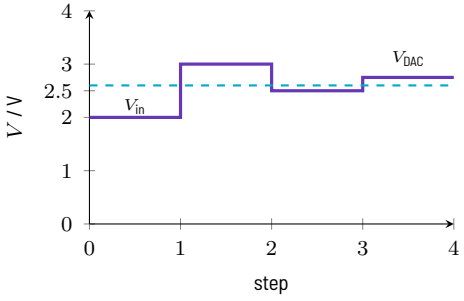
▶ Falstad: `examples/adc/r2r_dac.txt` Every node of the ladder sees a resistance of $2R$ to the left (the rest of the ladder) and $2R$ downwards (the bit input). Seen from each bit input, the current therefore splits in half at every node on its way to the output, so each bit has twice the weight of the next lower bit. A Thevenin conversion from left to right gives the formula. Loading the output with $4R$ instead of a follower scales it by $4R/(R + 4R) = 0.8$; this is used in the Falstad example `quant_error` to get exactly 0.5 V per code from 5 V logic levels.

Successive approximation (SAR)



- binary search with one comparator and a DAC: one bit per clock cycle, MSB first
- set the bit, compare V_{in} with V_{DAC}: keep the bit if $V_{in} \geq V_{DAC}$, otherwise clear it
- N steps for N bits; needs an S&H (built in); 8 …18 bit, up to some MS/s

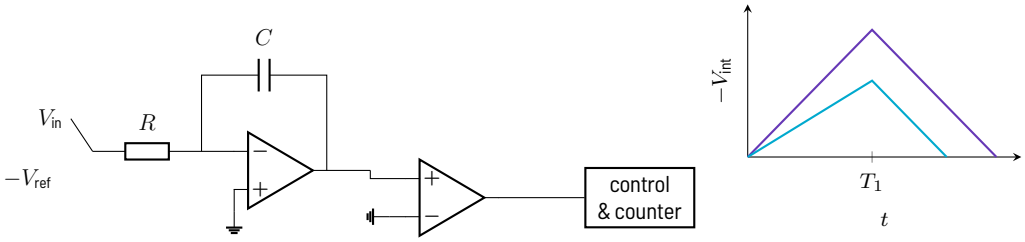
SAR: worked example



step	trial	V_{DAC}	$V_{in} \geq V_{DAC}$?
1	**1**000	2.00 V	yes: keep
2	1**1**00	3.00 V	no: clear
3	1**0**10	2.50 V	yes: keep
4	1**1**11	2.75 V	no: clear
	1010	2.50 V	$D = 10$

- 4 bit, $V_{FS} = 4$ V, LSB $= 0.25$ V, $V_{in} = 2.6$ V
- result 1010: V_{in} lies between 2.50 V and 2.75 V

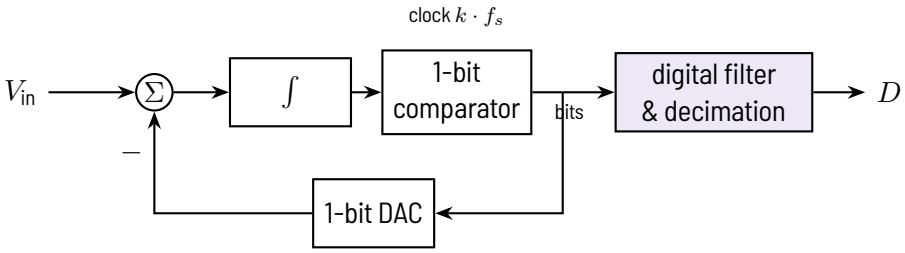
Dual-slope converter



- integrate V_{in} for a fixed time T_1, then integrate $-V_{ref}$ until the output is zero again (T_2)
- $V_{in}\,T_1 = V_{ref}\,T_2 \quad \Rightarrow \quad D = N_1 \dfrac{V_{in}}{V_{ref}}$: independent of R, C and the clock frequency
- $T_1 = 20$ ms (or 100 ms): 50 Hz interference integrates to zero

The dual-slope converter is slow (a few conversions per second) but very accurate, and it rejects mains interference by design: the integral of a 50 Hz sine over a whole number of periods is zero. That is why digital multimeters use it. It is the opposite strategy to our capstone, where the 50 Hz interference is removed by the anti-aliasing filter before sampling.

Sigma-delta converter (concept)



- oversampling with a 1-bit quantizer in a feedback loop: the density of ones is proportional to V_{in}
- the loop shapes the quantization noise to high frequencies; a digital low-pass removes it
- 16 …24 bit, audio and precision measurement (weighing scales, thermocouples)
- the digital filter is also the anti-aliasing filter: the analog filter can be simple

Comparison

	flash	SAR	dual-slope	sigma-delta
principle	$2^N - 1$ comparators	binary search	integration	oversampling
steps per sample	1	N	$2 \cdot 2^N$ clocks	many (oversampling)
resolution	4 …8 bit	8 …18 bit	12 …22 bit	16 …24 bit
speed	up to GS/s	kS/s …MS/s	1 …100 S/s	Hz …MHz bandwidth
cost, power	high (exponential)	low	low	medium
typical use	oscilloscopes, radio	microcontrollers	multimeters	audio, scales

- the capstone uses a 3-bit flash converter because every part of it is visible: ladder, comparators, logic

9.7 Exercises

Exercise 9.1: Aliasing

A system samples at $f_s = 5$ Hz.

a) Calculate the alias frequencies of input components at 2 Hz, 4 Hz, 12 Hz, 49 Hz and 50 Hz.

b) The sensor signal is 2 V DC with 0.5 V of 50 Hz hum. Describe the sampled values for $f_s = 5$ Hz and for $f_s = 4.9$ Hz.

c) An RC low-pass with $6.8\,\text{k}\Omega$ and $10\,\mu\text{F}$ is inserted before the S&H stage. Calculate f_c and the remaining hum amplitude. Compare with the LSB of the 3-bit converter (0.5 V).

d) Check part b) in Falstad.

Solution 9.1

a) 2 Hz: below $f_s/2$, no alias. 4 Hz: $|4-5| = 1$ Hz. 12 Hz: $|12-2\cdot5| = 2$ Hz. 49 Hz: $|49-10\cdot5| = 1$ Hz. 50 Hz: $|50 - 10\cdot5| = 0$ Hz.

b) $f_s = 5$ Hz: the hum is sampled at the same phase every time; all samples are equal, $2\,\text{V} + 0.5\,\text{V}\sin\varphi_0$, i.e. a constant offset between -0.5 V and 0.5 V. $f_s = 4.9$ Hz: the phase advances by 0.204 periods per sample; the samples follow a 1 Hz sine between 1.5 V and 2.5 V.

c) $f_c = 1/(2\pi\cdot 6.8\,\text{k}\Omega\cdot 10\,\mu\text{F}) = 2.34$ Hz. $|H(50\,\text{Hz})| = 1/\sqrt{1+(50/2.34)^2} = 0.047$ (-26.6 dB): the hum is reduced to 23 mV, less than $\frac{1}{20}$ LSB. The offset error disappears from the readings.

d) ▶ Falstad: `examples/adc/alias50.txt`
`Vs_5Hz` holds 2.14 V at every sample (offset 0.14 V for the phase in the example); `Vs_4.9Hz` shows 2.14 V, 2.50 V, 2.14 V, 1.58 V, 1.62 V, 2.20 V, …

Exercise 9.2: Quantization

A 3-bit converter has a full scale of 4 V and thresholds at $k\cdot$ LSB.

a) Calculate the LSB and the codes for 0.3 V, 1.8 V, 2.0 V and 3.98 V.

b) Which voltage range does the code 101 represent? What is the largest quantization error?

c) Calculate the SNR of the ideal 3-bit converter. How many bits are needed for an SNR of at least 60 dB?

d) Observe the quantization in Falstad.

Solution 9.2

a) LSB $= 0.5$ V. $D = \lfloor V_{\text{in}}/0.5\,\text{V}\rfloor$: 0.3 V $\to$ 000, 1.8 V $\to$ 011, 2.0 V $\to$ 011 or 100 (exactly on the threshold, the comparator offset decides), 3.98 V $\to$ 111.

b) 101 = 5: 2.5 V … 3.0 V. The error $V_{\text{in}} - D\cdot$ LSB is between 0 and 0.5 V (1 LSB); relative to the bin centre 2.75 V it is ± 0.25 V.

c) SNR $= 6.02\cdot 3 + 1.76 = 19.8$ dB. $6.02\,N + 1.76 \geq 60 \Rightarrow N \geq 9.7$: 10 bits.

d) ▶ Falstad: `examples/adc/quant_error.txt`
At $V_{\text{in}} = 1.8$ V the DAC output is $V_q = 1.5$ V, at 3.98 V it is 3.5 V; $V_{\text{in}} - V_q$ stays between 0 V and 0.5 V.

Exercise 9.3: Designing the 3-bit flash converter

Design the converter of the capstone: full scale $4.00\,\text{V}$, 3 bit, comparators $C_k = (V_{\text{in}} > k \cdot 0.5\,\text{V})$.

a) Dimension the reference ladder with $1\,\text{k}\Omega$ resistors. Calculate the tap voltages, the ladder current and the power taken from the reference.

b) The comparators have an input bias current of $100\,\text{nA}$. Estimate the worst-case threshold shift. (Hint: Thevenin resistance of a tap.)

c) Write down the truth table from thermometer code to binary and derive the logic equations for B_2, B_1, B_0.

d) How must the ladder be changed for thresholds at $(k - \frac{1}{2}) \cdot \text{LSB}$?

e) Check the converter in Falstad at the bin centres.

Solution 9.3

a) 8 resistors of $1\,\text{k}\Omega$ from $4\,\text{V}$ to ground. Taps $V_k = k \cdot 4\,\text{V}/8 = 0.5\,\text{V}, 1.0\,\text{V}, \ldots, 3.5\,\text{V}$. $I = 4\,\text{V}/8\,\text{k}\Omega = 0.5\,\text{mA}$, $P = 2\,\text{mW}$.

b) The largest Thevenin resistance is at the middle tap: $4\,\text{k}\Omega \parallel 4\,\text{k}\Omega = 2\,\text{k}\Omega$. Shift: $100\,\text{nA} \cdot 2\,\text{k}\Omega = 0.2\,\text{mV}$, negligible compared with $0.5\,\text{V}$.

c) Truth table as on the slide *Thermometer code*. With $H_k = C_k\overline{C_{k+1}}$: $B_2 = H_4 + H_5 + H_6 + H_7 = C_4$; $B_1 = H_2 + H_3 + H_6 + H_7 = C_2\overline{C_4} + C_6$; $B_0 = H_1 + H_3 + H_5 + H_7 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7$.

d) Total resistance stays $8R$: bottom resistor $R/2 = 500\,\Omega$, six resistors R, top resistor $3R/2 = 1.5\,\text{k}\Omega$. The taps are then $0.25\,\text{V}$, $0.75\,\text{V}$, ..., $3.25\,\text{V}$.

e) ▶ Falstad: `examples/adc/flash3.txt`
Taps $0.5\,\text{V}$ ...$3.5\,\text{V}$, ladder current $0.50\,\text{mA}$. Slider at $0.25\,\text{V}$, $0.75\,\text{V}$, ..., $3.75\,\text{V}$: the bar graph shows 0 ...7 LEDs, the binary LEDs 000, 001, ..., 111.

Exercise 9.4: Bubble error

Because of noise, the comparators of the 3-bit flash converter deliver the pattern 0010111 (C_7 first), a bubble at C_4.

a) Which input range is plausible? Which code would be correct?

b) Which code does the encoder of the capstone produce?

c) Which code does a priority encoder produce?

Solution 9.4

a) $C_5 = 1$ but $C_4 = 0$ contradict each other. Either C_4 is wrong (input between $2.5\,\text{V}$ and $3.0\,\text{V}$, code 5) or C_5 is wrong (input between $1.5\,\text{V}$ and $2.0\,\text{V}$, code 3).

b) $B_2 = C_4 = 0$; $B_1 = C_2\overline{C_4} + C_6 = 1$; $B_0 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7 = 0 + 1 + 1 + 0 = 1$: code 011 = 3, one of the plausible values. (For the pattern 0000101 of the slide, however, it gives 001 instead of 011.)

c) The priority encoder looks at the highest active input, C_5: code 101 = 5, the other plausible value. For 0000101 it gives 011 = 3, the correct value; it never produces a code below the top of the column.

Exercise 9.5: LED display

The bar graph and the binary LEDs are driven by $5\,\text{V}$ logic outputs. Red LEDs with $V_F \approx 2\,\text{V}$ shall carry $5\,\text{mA} \ldots 10\,\text{mA}$.

a) Choose the series resistor from the E12 series and calculate the LED current.

b) Calculate the total LED current for the code 111.

c) In the Falstad model, C_4 drives its bar LED and also the B_2 LED. Which current does this output deliver?

d) Why must each LED have its own resistor?

Solution 9.5

a) $R = (5\,\text{V} - 2\,\text{V})/I_F$ gives $300\,\Omega \ldots 600\,\Omega$. Choose $330\,\Omega$: $I_F = 3\,\text{V}/330\,\Omega = 9.1\,\text{mA}$.

▶ Falstad: `examples/adc/flash3.txt`

The Falstad LED has $V_F = 1.77\,\text{V}$, so the current is $3.23\,\text{V}/330\,\Omega = 9.8\,\text{mA}$.

b) 10 LEDs: $10 \cdot 9.1\,\text{mA} = 91\,\text{mA}$.

c) $2 \cdot 9.1\,\text{mA} = 18\,\text{mA}$ (plus negligible gate input currents). That is near the limit of a CMOS logic output; a buffer or lower LED currents ($680\,\Omega$: $4.4\,\text{mA}$) are safer.

d) LEDs of different colours and samples have different V_F. With a common resistor the LED with the lowest V_F takes most of the current, and the current per LED changes with the number of LEDs that are on.

Exercise 9.6: Schmitt trigger, R-2R DAC and SAR

a) A non-inverting Schmitt trigger has $V_{\text{ref}} = 2\,\text{V}$, $R_1 = 10\,\text{k}\Omega$ and output levels $0\,\text{V}/5\,\text{V}$. Choose R_2 for a hysteresis of $0.5\,\text{V}$ and calculate both thresholds.

b) A 3-bit R-2R DAC is driven by $5\,\text{V}$ logic levels. Calculate the step size and the output for $D = 5$.

c) A 4-bit SAR converter ($V_{\text{FS}} = 4\,\text{V}$) converts $V_{\text{in}} = 1.3\,\text{V}$. Write down the trial codes, the DAC voltages and the result.

d) Check a) and b) in Falstad.

Solution 9.6

a) Hysteresis $V_{0\text{H}} R_1/R_2 = 0.5\,\text{V} \Rightarrow R_2 = 5\,\text{V} \cdot 10\,\text{k}\Omega/0.5\,\text{V} = 100\,\text{k}\Omega$. $V_{T+} = 2\,\text{V} \cdot 1.1 = 2.2\,\text{V}$, $V_{T-} = 1.7\,\text{V}$.

b) LSB $= 5\,\text{V}/8 = 0.625\,\text{V}$; $D = 5$: $3.125\,\text{V}$.

c) LSB $= 0.25\,\text{V}$. 1000: $2.00\,\text{V}$, no $\rightarrow$ 0. 0100: $1.00\,\text{V}$, yes $\rightarrow$ 1. 0110: $1.50\,\text{V}$, no $\rightarrow$ 0. 0101: $1.25\,\text{V}$, yes $\rightarrow$ 1. Result 0101 = 5 ($1.25\,\text{V} \ldots 1.50\,\text{V}$).

d) ▶ Falstad: `examples/adc/comparator_hyst.txt`

The Schmitt trigger switches at $2.2\,\text{V}$ (rising) and $1.7\,\text{V}$ (falling) and gives exactly two edges per period; the plain comparator produces 10 edges.

▶ Falstad: `examples/adc/r2r_dac.txt`

`Vout` steps through $0\,\text{V}$, $0.625\,\text{V}$, ..., $4.375\,\text{V}$; at $D = 5$ it shows $3.125\,\text{V}$.

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